



Innovative Applications of O.R.

Benchmarking culture in Europe: A data envelopment analysis approach to identify city-specific strengths

Tom Van Puyenbroeck^a, Valentina Montalto^{b,*}, Michaela Saisana^b^a Faculty of Economics and Business, KU Leuven, Warmoesberg 26, 1000 Brussels, Belgium^b European Commission, Joint Research Centre, Via E. Fermi, 2749, 21027, Ispra

ARTICLE INFO

Article history:

Received 18 March 2019

Accepted 29 May 2020

Available online 6 June 2020

Keywords:

Composite indicator(s)

Creative city

Data envelopment analysis

Urban policy

ABSTRACT

Culture is an integral part of a city's quality of life, a driver of urban change, and a genuine economic sector. To support benchmarking of urban culture and facilitate peer learning amongst policymakers, the European Commission has recently created the 'Cultural and Creative Cities Index'. While this index builds on a standardised method to aggregate 29 indicators for the 155 selected cities, it is explicitly acknowledged that a 'gold standard' for a 'Cultural and Creative City' does not exist. Instead, different approaches should be allowed for in capturing cities' cultural and creative vitality. This is the point of departure for this paper, which employs a Benefit-of-the-Doubt (BoD) modelling approach to allow cities to combine such respect for performance diversity with peer learning and benchmarking. Expert-based weights are used to provide expert-consistent bounds for the shares of key dimensions in a city's final BoD index value. We identify three city clusters, amongst which there are large performance differences. Accordingly, we focus on the within-group identification of peer cities and target values, which we illustrate in more detail for Bilbao, Kraków and Umeå, towards the formulation of fit-for-purpose policy measures that can support culture-led development.

© 2020 The Author(s). Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license.

<http://creativecommons.org/licenses/by-nc-nd/4.0/>

1. Introduction

Culture is increasingly seen as an economic sector of its own, contributing to growth and job creation (KEA, 2006; Throsby, 2001; Unctad, 2010) as well as fostering creativity and innovation (Bakhshi, MacVittie & Simmie, 2008; Pratt & Jeffcutt, 2009a). Culture is also considered to be a major vehicle for urban change, contributing to cities' regeneration, social inclusion and wellbeing (Blessi, Grossi, Sacco, Pieretti & Ferilli, 2016; Evans & Shaw, 2004; Ferilli, Sacco, Tavano Blessi & Forbici, 2017).

Culture has thus emerged as a crucial policy dimension when addressing issues such as attractiveness, innovation and social cohesion needs. This is true at all spatial levels of policy intervention: from the city level (Miles & Paddison, 2005) right up to the level of countries or supranational groups of countries like the European

Union (EU) (e.g. European Commission, 2010a, 2010b, 2010c, 2012, 2014a, 2014b, 2016).

While this new understanding of culture has been introduced into the EU policy agenda and into the policy agendas of many of its Member States, regions and cities, the practical implementation of culture-led development strategies remains a challenge. Amongst other reasons, this is related to the fact that culture-led development is multidimensional, covering different domains of the economy, society and individuals' lives. Culture-led development requires a comprehensive policy supported by a sound evidence base capturing the diverse cultural resources that can be mobilised for development purposes.

The Cultural and Creative Cities Index ('C3 Index') is a composite indicator that aggregates information regarding culture – covering the aspects 'Cultural Vibrancy', 'Creative Economy' and 'Enabling Environment' – for 155 cities in the EU-28 countries. These cities were selected for inclusion in the index thanks to their active engagement in promoting culture and creativity, demonstrated by having been, about to become, or shortlisted to become European Capitals of Culture; being part of the UNESCO Creative Cities network; or hosting at least two regular international cultural

* Corresponding author.

E-mail addresses: tom.vanpuyenbroeck@kuleuven.be (T. Van Puyenbroeck), valentina.montalto@ec.europa.eu (V. Montalto), michaela.saisana@ec.europa.eu (M. Saisana).

festivals. The C3 Index was created by the European Commission's Directorate General Joint Research Centre (DG JRC) as part of the Cultural and Creative Cities Monitor (CCCM) project, which provides an aggregate C3 Index score, but also allows for benchmarking on the underlying 29 indicators, nine policy dimensions and three sub-indices. As an aggregate measure, the total score of the C3 Index for each city is calculated based on a fixed set of weights, which has been defined by a panel of experts specialised in culture, creativity and urban development. Cities have consequently been ranked based on their C3 Index scores, ranging from 0 to 100, with higher scores signalling better performance. The maximum C3 Index value is to be regarded as a theoretical possibility rather than a realistic target. Paris, for instance, ranks first out of the 155 ranked EU cities, with a score of 63.2, showing that there is potential for improvement for all urban areas.

The actual development of culture-led policies at the city level, however, entails an intricate balance between international policy priorities and city-specific conditions. The CCCM fully acknowledges that a universal blueprint for a 'Cultural and Creative City' does not exist; diversity itself provides a learning platform for cities wishing to further develop a culturally vibrant ecosystem. To that end, city data are primarily compared with those for similar peer cities, based on population size, GDP per capita, or employment rates.

Nevertheless, context-specific solutions may not be easily identified when peer learning and benchmarking rely on an index with fixed weights. The choice of weights is indeed a crucial step in the development of a composite performance measure, which has become an increasingly popular tool for assessing and ranking entities (e.g. countries, institutions) on complex concepts that are not directly measurable (the literature on the topic is very rich, for a more comprehensive review, see e.g. Saisana, Saltelli & Tarantola, 2005; Saltelli, 2007; as well as more recent works e.g. Brousmiche et al., 2020; El Gibari, Gómez & Ruiz, 2019; Gasser, 2020; Kwatra, Kumar & Sharma, 2020). The JRC-OECD Handbook (2008) highlights two main implications for the weights: (1) a potentially significant impact on the scores and rankings, and (2) potential to influence the behaviour of the units evaluated, as improvement efforts will likely be focused on the indicators/dimensions that are more heavily weighted. Furthermore, comparing cities' multidimensional performance through the lens of a fixed and common set of weights may even be counterproductive in that it prevents acceptance of such a benchmarking tool.

In this paper, we discuss the use of a composite indicator construction methodology that inherently accounts for the diversity of perspectives, by giving cities considerable 'Benefit-of-the-Doubt' (BoD) in the choice of weights (see e.g., Cherchye, Moesen, Rogge & Puyenbroeck, 2007; 2008). This approach is based on the more broadly known Data Envelopment Analysis (DEA) that has been applied in a variety of settings. The BoD approach also helps in identifying relative strengths, areas of improvement and achievable targets. Furthermore, its use can enhance the acceptability of an index as a tool for guiding the formulation of fit-for-purpose policies.

We use the nine key dimensions of the C3 Index to calculate city-specific weights and to further aggregate the nine dimensions into an overall composite indicator of culture for European cities. To do so, we apply an optimisation procedure constrained by the minimum and maximum values of the 'importance shares' of the nine dimensions across the 155 cities. One novel aspect of our approach is that, contrary to the usual case in BoD applications, we combine city-specific dimension scores with (fixed) 'pure weight' information provided by experts to identify these 'importance share' bounds.

Technically, such bounds boil down to 'proportional virtual weight restrictions' (Wong & Beasley, 1990). We first follow the majority of BoD (and DEA) papers by applying such restrictions to

the unit evaluated only (in our case, the city). However, Sarrico and Dyson (2004) demonstrated that this may lead to optimal weights that might generate counterintuitive results, when applied to the data that constitute the possible peers of the unit assessed. In fact, this may even lead to identifying specific benchmark units which themselves violate the restrictions imposed on the unit assessed. Departing from most of the existing literature, we explicitly address this inconsistency problem here and check for such violations in our empirical application. A second novel aspect of this paper is that we apply a heuristic approach as a remedy: we first detect all observations that are identified as benchmarks for other cities, and then append the BoD optimisation problem with additional restrictions to ensure that such 'potential' benchmarks effectively behave in the desired manner.

In the next section we detail the methodological choices underlying the original construction of the C3 Index. In the third section, we discuss our alternative methodology to measure culture in its multidimensional nature in European cities. In the fourth section, various results are presented and discussed, both for the entire 155 city group as well as for a subsample of 'similar' cities (based on cluster analysis). We analyse *ex post* whether and to what extent important group differences exist in aggregate cultural and creative best practices. Our empirical analyses reveal that this is indeed the case. We further include some illustrative cases to provide policy directions for specific cities. The fifth section presents concluding remarks.

2. The Cultural and Creative Cities Index

The C3 Index is a composite indicator. Composite indicators are indices that aggregate several indicators of performance to assess multidimensional concepts. In other words, they can provide policymakers with a summary measure of complex socio-economic phenomena, facilitating comparisons and benchmarking across countries, regions or cities, and monitoring progress over time. The development and use of sound and relevant composite indicators however requires a number of subtle decisions to be made, ranging from concept definition to selection of indicators to the choice of the most appropriate weighting scheme, amongst others. For the C3 Index, the ten-step guide jointly developed by JRC and OECD in their *Handbook on constructing composite indicators* (JRC-OECD, 2008) was followed; this proposes the definition of a clear theoretical framework as a first main step.

The C3 Index fits into the research path recognising culture as an eminently urban phenomenon and a 'resource' for urban development (Montalto, Tacao Moura, Langedijk & Saisana, 2019). Culture and cities can be thought of as natural counterparts. Most of Europe's cultural heritage is found in cities. The urban environment is thus a privileged context for cultural facilities, activities and participation. Conversely, culture has important cumulative effects on urban quality of life and well-being (Ferilli et al., 2017; Prior & Blessi, 2012).

Today, cultural industries also fit naturally with the requirements of the knowledge economy. The creative knowledge typical of art and culture has important spill-over effects to other information-intensive economic sectors, ranging from cultural tourism (OECD, 2009) to consumer electronics (Bakhshi et al., 2008; Potts, 2009; Pratt & Jeffcutt, 2009b).

By gathering together academic, research and cultural centres, technological infrastructures, efficient laws and regulations as well as media and communication opportunities, cities offer the most favourable conditions to put '[cultural] resources to good use' (UNCTAD, 2013, p. 126) and stimulate creative economies (World Economic Forum, 2016), supporting both economic and social health.

Table 1
The Cultural and Creative Cities' Monitor 2017: conceptual framework, indicators and expert-based weights.

Sub-index Weights(1)	Sub-index	Dimension Weights (2)	Dimensions	Rescaled Dimension Weights= (1) x (2)	Indicators
40%	1. Cultural Vibrancy	50%	D1.1 Cultural Venues & Facilities	20%	1. Sights & landmarks 2. Museums 3. Cinema seats 4. Concerts & shows 5. Theatres
		50%	D1.2 Cultural Participation & Attractiveness	20%	6. Tourist overnight stays 7. Museum visitors 8. Cinema attendance 9. Satisfaction with cultural facilities
40%	2. Creative Economy	40%	D2.1 Creative & Knowledge-based Jobs	16%	10. Jobs in arts, culture & entertainment 11. Jobs in media & communication 12. Jobs in other creative sectors
		20%	D2.2 Intellectual Property & Innovation	8%	13. ICT patent applications 14. Community design applications
		40%	D2.3 New Jobs in Creative Sectors	16%	15. Jobs in new arts, culture & entertainment enterprises 16. Jobs in new media & communication enterprises 17. Jobs in new enterprises in other creative sectors
20%	3. Enabling Environment	40%	D3.1 Human Capital & Education	8%	18. Graduates in arts and humanities 19. Graduates in ICT 20. Average appearances in university rankings
		40%	D3.2 Openness, Tolerance & Trust	8%	21. Foreign graduates 22. Foreign-born population 23. Tolerance of foreigners 24. Integration of foreigners 25. People trust
		15%	D3.3 Local & International Connections	3%	26. Passenger flights 27. Potential road accessibility
		5%	D3.4 Quality of Governance	1%	28. Direct trains to other cities 29. Quality of governance

Still, not all cities are the same, just as culture itself is a phenomenon that tends to have at least some intensely local characteristics (Santagata, 2002; Scott, 2001; van Oort, 2008) helping to differentiate places from one another.

The C3 Index was created with the purpose of enabling policymakers to benchmark three key measurable manifestations of urban culture, to facilitate comparisons, peer learning and policy action.

- As the ensemble of cultural facilities, activities and participation have important effects on urban identities and quality of life, the ‘Cultural Vibrancy’ domain aims to measure the presence of a diverse set of cultural venues as cultural participation opportunities, and the capacity of such venues to attract audiences.
- As the links between local culture and urban economic activities have become increasingly evident, the ‘Creative Economy’ domain intends to capture how well culture contributes to a city’s economy, and more specifically to jobs and job creation in cultural and creative sectors (CCS) and to innovation outputs.
- As cities are expected to offer favourable conditions for the development of lively cultural environments and creative economies, the ‘Enabling Environment’ domain seeks to capture at least some of these conditions, such as the presence of high quality universities, transport connections, a climate of openness, tolerance and trust, and an efficient governance system.

In order to benchmark the performance of EU cities in these three areas, a set of indicators was selected for each domain. These indicators were chosen based on their scientific relevance to the concept, timeliness, public availability and quality of data sources, and best data coverage for the 155 selected EU cities. Furthermore, in order to produce an overall aggregate score for each city, the developers of the C3 Index consulted a group of experts, which included policymakers, academics and practitioners working on culture and local development at the international level. The

weights were obtained using the Budget Allocation method (JRC-OECD, 2008) in which individuals were requested to allocate points in two steps: first to the three overarching domains and then to the dimensions within each domain, with more points assigned to the more important components. However, the weights so obtained should be multiplied by the dimensions score for a correct interpretation of their contribution to the index score, as further discussed in Section 3.

Table 1 presents the complete list of 29 indicators, their relation to the nine dimensions and the three overarching sub-indices obtained for the three domains presented above, as well as the fixed weights attributed by the experts to sub-indices and to dimensions within each of the sub-indices. For later reference, we have also included rescaled dimension weights (in *italics*).¹

It can be seen that the experts agreed on a set of weights reflecting the primary importance of ‘Cultural Vibrancy’ and the related ‘Creative Economy’ domain. This set of weights is intended to stimulate greater investments in culture and creativity as genuine engines of sustainable development and growth, rather than in complementary enablers such as transport connections or the governance system. In addition, as agreed by the index developers together with the experts consulted, weights were not applied at the ‘granular’ level of the indicators: the three domains and nine dimensions were considered as more pertinent to capture concepts of major policy interest (e.g. cultural vibrancy). Of course, the indicator level remains highly relevant when it comes to deciding on specific policy actions. To improve cultural vibrancy, for instance, a city may decide to invest in museum engagement policies (to improve the museum visitors indicator) or rather in the number and variety of the local cultural infrastructure (to improve the theatre

¹ For a more detailed description of the indicators and the steps leading to construction of the C3 Index, see Montalto, Tacao Moura, Langedijk, & Saisana, 2017 and the accompanying Annex A: The Cultural and Creative Cities Monitor methodology in ten steps available at: <http://publications.jrc.ec.europa.eu/repository/bitstream/JRC107331/kj0218784enn.pdf>

Table 2

City peer groups in the cultural and creative cities monitor.

GDP per capita groups (in PPS)	Number of cities	Employment rate groups	Number of cities	Population groups	Number of cities
> 35,000	32	> 74%	40	> 1 million (XXL)	21
30,000–35,000	26	71–74%	22	500,000 - 1,000,000 (XL)	34
25,000–30,000	36	68–71%	23	250,000 - 500,000 (L)	36
20,000–25,000	24	65–68%	28	100,000 - 250,000 (M)	47
< 20,000	37	< 65%	42	50,000 - 100,000 (S)	17
TOT	155		155		155

Table 3

C3 Index – 2017 edition: Top five cities per population group.

C3 Index Rank		Sub-index Ranks		
#	City-country	1. Cultural Vibrancy	2. Creative Economy	3. Enabling Environment
[XXL group] > 1,000,000 inhabitants (21 cities)				
1	Paris-FR	1	1	2
2	Munich-DE	9	2	3
3	Prague-CZ	2	6	16
4	Milan-IT	3	11	8
5	Brussels-BE	10	3	13
[XL group] 500,000–1,000,000 inhabitants (34 cities)				
1	Copenhagen-DK	2	2	5
2	Amsterdam-NL	4	3	7
3	Lisbon-PT	1	9	15
4	Stockholm-SE	5	6	4
5	Dublin-IE	3	10	1
[L group] 250,000–500,000 inhabitants (36 cities)				
1	Edinburgh-UK	5	8	3
2	Karlsruhe-DE	21	2	4
3	Utrecht-NL	11	4	1
4	Nuremberg-DE	13	3	8
5	Florence-IT	1	21	19
[M group] 100,000–250,000 inhabitants (47 cities)				
1	Eindhoven-NL	3	2	9
2	Linz-AT	4	4	11
3	's-Hertogenbosch-NL	2	7	10
4	Cork-IE	1	18	8
5	Heidelberg-DE	10	5	1
[S group] 50,000–100,000 inhabitants (17 cities)				
1	Galway-IE	2	6	2
2	Leuven-BE	14	1	1
3	Weimar-DE	1	2	8
4	Limerick-IE	3	8	3
5	Waterford-IE	4	5	6

or concerts halls indicator), depending on the local context and priorities.

Meaningful benchmarking is made possible in the CCCM by dividing cities into five predefined peer groups, based on similar population, annual GDP per capita, or employment rate. In other words, the C3 Index deliberately avoids providing a general comparison amongst the 155 cities, as there are huge variations in their scale. Instead, multiple city rankings aim to reflect and respect the diversity of European urban settings, as well as to facilitate peer learning from cities that share similar demographic or socioeconomic characteristics.² Table 2 details how the 155 cities are grouped.

Table 3 presents the top five cities on the C3 Index with respect to the five population size groups³ and based on the scores obtained with the fixed set of weights presented above.

Paris, Copenhagen, Edinburgh, Eindhoven and Galway rank first in their respective population groups. Interestingly, however, top cities do not necessarily excel in all dimensions, with Paris being a notable exception. Copenhagen, for instance, takes first place amongst cities of 500,000 to 1,000,000 inhabitants (XL cities), ranking second on both 'Cultural Vibrancy' and 'Creative Economy' and fifth on 'Enabling Environment'. Edinburgh ranks first amongst cities of 250,000 to 500,000 inhabitants (L cities), taking fifth position on 'Cultural Vibrancy', eighth on 'Creative Economy' and third on 'Enabling Environment'. Eindhoven leads amongst medium-sized cities of 100,000 to 250,000 inhabitants (M cities), ranking third on 'Cultural Vibrancy', second on 'Creative Economy' and ninth on 'Enabling Environment'.⁴

² See the 'Compare my city with...' interactive feature on each city page of the accompanying online tool: <https://composite-indicators.jrc.ec.europa.eu/cultural-creative-cities-monitor/>.

³ The full rankings by income, employment and population groups can be found on the CCCM's online tool: <https://composite-indicators.jrc.ec.europa.eu/cultural-creative-cities-monitor/>.

⁴ The cities' position in the three sub-indices is explained by different performance on the underlying indicators. Within the 'Cultural Vibrancy' sub-index, for instance, Copenhagen leads on Theatres and comes third on Sights & landmarks, Cinema attendance and Museum visitors. Edinburgh ranks fourth on three indicators (Sights & landmarks, Museums, and Concerts & shows), whilst Eindhoven is first on Cinema seats and second on Theatre. Within 'Creative Economy', Paris leads on five indicators related to jobs and job creation in the cultural and creative sectors, and on the two indicators related to innovation outputs; Copenhagen is first on Jobs in arts, culture & entertainment and Jobs in new media & communication enterprises and second on Jobs in media & communication; while Eindhoven leads on

Despite an evident respect for diversity in its final presentation, which is itself grounded in the fundamental idea that there are different ways in which a city can excel in this area, there is one aspect of the (standardised) C3 Index that is at odds with this consideration, namely the use of a uniform weighting scheme common to all cities. It was already noted in the introduction that, in the specific setting at hand, it is hard to believe that the weights attached to the index components should be the same for all cities. Put differently, in this context the use of city-specific weights has the two main advantages of, first, further strengthening the policy relevance and use of an index, and, second, facilitating the identification of realistic benchmarks and targets. In the next section, we discuss how the use of a DEA-based methodology can naturally deal with these aspects.

3. An alternative approach that respects diversity in measuring culture in European cities

Flexible weighting is a natural outcome of DEA models, as weights are derived endogenously. When used for constructing a composite indicator, this characteristic implies that city-specific weights are derived on the basis of a linear programme, such that a city's composite indicator value, i.e. a weighted sum of its indicator values, is maximised relative to the similarly weighted aggregates of all other cities. This approach is often termed as Benefit-of-the-Doubt (BoD) weighting (e.g., [Cherchye et al., 2007; 2008](#)).

BoD weighting thus crucially involves relative comparisons. Also, the concept of benchmarking fits naturally with the BoD approach, as a city's benchmarks are those other cities that yield the highest possible composite indicator value achievable with the assessed city's (most favourable) weights.

Let us formalise this intuition. Consider the nine dimensions covered by the CCCM for $n = 155$ cities, with y_{ij} being the observed value of dimension i in city j (thus, $j = 1, \dots, 155$). Assume we are interested in assessing city 'o'. Let $w_{io} \geq 0$ represent the weight of the i th dimension ($i = 1, \dots, 9$) in this city 'o'. One possibility in making relative comparisons is to apply city o's weights also to the other cities' data and to look at the n fractions:

$$\frac{\sum_{i=1}^9 w_{io} y_{ij}}{\sum_{i=1}^9 w_{io} y_{io}}, \quad j = 1, \dots, 155.$$

Clearly, city 'o' is outperformed by another city whenever the numerator exceeds the denominator. Benchmark cities for 'o' are actually those that, given the weights w_{io} , yield the highest aggregate score, i.e.

$$\max_{j=1, \dots, 155} \frac{\sum_{i=1}^9 w_{io} y_{ij}}{\sum_{i=1}^9 w_{io} y_{io}}$$

The idea of BoD weighting then entails looking for weights that are most favourable for city 'o', i.e. weights that are consistent with:

$$\min_{w_{io}} \frac{\max_j \sum_{i=1}^9 w_{io} y_{ij}}{\sum_{i=1}^9 w_{io} y_{io}}. \quad (1)$$

three indicators - Jobs in other creative sectors, ICT patent applications and Community design applications - scoring particularly well on patent and new design applications.

This is a fractional optimisation problem, which can be turned into a linear programming model by adding a normalisation constraint, ensuring that the denominator in the above expression is 1. One consequently obtains:

(P1):

min Φ

s.t.

$$\sum_{i=1}^9 w_{io} y_{io} = 1, \quad (2)$$

$$\sum_{i=1}^9 w_{io} y_{ij} \leq \Phi, \quad j = 1, \dots, 155 \quad (3)$$

$$w_{io} \geq 0, \quad i = 1, \dots, 9 \quad (4)$$

Note that, as the subscript o refers to the one city that is assessed, (P1) must be solved 155 times. In each case Φ can be readily interpreted as the degree by which a benchmark city outperforms the assessed city, with $\Phi \geq 1$. A Φ -value of 1 conveys that no other city exists which outperforms the assessed city, given the latter's optimal weights. Below, we will mostly stick to the convention to report $0 < 1/\Phi \leq 1$ as a city's actual BoD-score, such that higher values indicate better overall performance.

Apart from the non-negativity requirements (4) in (P1), there is total flexibility in the choice of weights. There are two reasons why we opt for additional constraints in our actual model. First, leaving out such additions could lead to an endogenous choice of zero weights applying to some of the nine domains, thus compromising the actual CCCM's holistic perspective on culture. In addition, zero weights would strongly contrast with the expert consensus opinion as summarised in [Table 1](#). Secondly, we will in fact directly build on this consensus opinion to narrow down the range of possible weight values. Indeed, the fixed expert weights can be reconciled with the idea of (restricted) BoD weighting.

Of course, in a direct reading these two features are plainly irreconcilable. However, it has been pointed out earlier in the composite indicator literature that 'weights' should not be regarded as a direct measure of the 'importance' of an index component for the resulting composite indicator value ([Becker, Saisana, Paruolo & Vandecasteele, 2017; JRC-OECD, 2008; Paruolo, Saisana & Saltelli, 2013](#)). A similar distinction is found in the DEA literature, which differentiates between the 'pure weights' w_{io} – that essentially provide information on the trade-offs between different dimensions – and 'virtual outputs' $w_{io} y_{io}$, i.e. the product of a weight and its corresponding indicator value, which 'have a more clear interpretation in terms of the importance attached to a given variable by the unit under assessment' ([Cooper, Ruiz & Sirvent, 2011, p. 102](#)). It is this distinction that we build on next.

Specifically, consider experts-based dimension weights as given in [Table 1](#), and let these be denoted by W_i ($i = 1, \dots, 9$). It is straightforward that multiplying these fixed weights with city-specific y_{io} values reveals differences between cities regarding the importance of a specific dimension in terms of its contribution to their final index values. Formally, if we capture importance by a dimension's percentage contribution to the total index score for a given city, i.e.

$$\frac{W_i y_{ij}}{\sum_{i=1}^9 W_i y_{ij}}, \quad (5)$$

it is clear that a policy dimension can take on different importance values for different cities, despite the existence of fixed dimension weights.⁶

⁶ For example, almost a quarter (23%) of Berlin's C3 Index score of 34.6 rests on its performance on the dimension Cultural Participation & Attractiveness, 23%

Table 4

Lower and upper bounds for the share of the CCCM 2017 nine dimensions across 155 cities.

	Lower bound for the share of the CCCM dimension	Upper bound for the share of the CCCM dimension	Average importance share
D1.1 Cultural Venues & Facilities	6.4%	42.2%	19.9%
D1.2 Cultural Participation & Attractiveness	7.2%	35.2%	21.0%
D2.1 Creative & Knowledge-based Jobs	3.1%	33.1%	16.3%
D2.2 Intellectual Property & Innovation	0.7%	20.4%	5.3%
D2.3 New Jobs in Creative Sectors	0.6%	45.8%	14.7%
D3.1 Human Capital & Education	0.4%	21.3%	7.1%
D3.2 Openness, Tolerance & Trust	3.5%	23.7%	9.3%
D3.3 Local & International Connections	0.4%	7.5%	4.0%
D3.4 Quality of Governance	0.046%	4.4%	2.3%

We exploit the latter feature upon appending programme (P1) with 'proportional virtual output restrictions' (Wong & Beasley, 1990):

$$L_i \leq \frac{w_{io}y_{io}}{\sum_{i=1}^9 w_{io}y_{io}} \leq U_i, \quad i = 1, \dots, 9 \quad (6)$$

which, given the above, can be regarded as dimension importance bounds on a city's (BoD-weighted) index score. For each of the nine dimensions, the lower (L_i) and upper (U_i) bounds are determined from 6 and the actual dataset: we identify the minimum and maximum importance values consistent with the expert weights across all 155 cities. For instance, for dimension D1.1, Cultural Venues & Facilities, these implied importance shares ranging from 6.45% in Munich to 42.19% in Matera; for the dimension D3.4, Quality of Governance, these shares range from 0.046% in Sofia to 4.411% in Saint-Etienne, and so on. Table 4 displays the values of the resulting bounds, as well as information about the average importance shares, for all dimensions.

It should be noted that, while weight restrictions of the form (6) have been used before in several BoD applications, information about expert opinion (on weights) is usually *directly* taken as an indicator of a dimension's 'importance' and subsequently used to define lower and/or upper bounds.⁷ But, as noted, there is a subtle but real difference between these two concepts, and it seems advisable to preserve and exploit that difference in our application as well as in similar ones.⁸

Furthermore, just as almost any other BoD paper with weight restrictions as given in (6), the current one proposes applying these to the evaluated city only. As already shown by Sarrico and Dyson (2004), such an asymmetric treatment might entail that other units may well violate the bounds when assessing the efficiency of a specific unit. That is, while optimal weights in the appended version of (P1) will be consistent with both (6) and the comparison constraints (3) by construction, it is still possible that $L_i > \frac{w_{io}y_{ij}}{\sum_{i=1}^9 w_{io}y_{ij}}$ or $\frac{w_{io}y_{ij}}{\sum_{i=1}^9 w_{io}y_{ij}} > U_i$ for $j \neq o$.⁹ To the best of our

on New Jobs in Creative Sectors, 18% on Creative & Knowledge-based Jobs, 10% on Cultural Venues & Facilities, and the remaining 26% from the other five dimensions. Using the same fixed weights, the city of Ghent has a comparable C3 Index score of 33.0, but 33% of this value is due to the contribution of the Cultural Venues & Facilities dimension, 22% to Cultural Participation & Attractiveness, 14% to Creative & Knowledge-based Jobs, and only 2% to New Jobs in Creative Sectors.

⁷ This is the case in, for example, Cherchye et al. (2007), where information from different experts' weighting schemes is directly translated into importance bounds.

⁸ There is as such no guarantee that experts themselves, when providing their consensus weights, make a clear distinction between 'pure weights' and 'importance shares'. The rightmost column of Table 4 shows however that average importance shares, as calculated over all cities, are considerably close to the consensus weights reported in Table 1. In other words, at least in our specific application, even when experts themselves would confuse their consensus weighting scheme with a vector of importance weights, our results would still largely be consistent with such a position.

⁹ Zanella, Camanho, & Dias (2015) provide an exception to this standard approach by applying proportional virtual weight restrictions to an 'average' (artificial) obser-

knowledge, empirical applications of the BoD method have largely sidestepped this issue. We will however return to it explicitly in the next section.

In the next section, we will also make use of the dual formulation of (P1), including additional restrictions stemming from (6), namely:

(P2)

$$\max \varphi + \sum_{i=1}^9 \alpha_i L_i - \sum_{i=1}^9 \beta_i U_i \quad (7)$$

s.t.

$$\sum_{j=1}^{155} \lambda_j y_{ij} \geq (\varphi + \alpha_i - \beta_i) y_{io}, \quad i = 1, \dots, 9 \quad (8)$$

$$\sum_{j=1}^{155} \lambda_j = 1, \quad \lambda_j \geq 0. \quad (9)$$

The α_i 's and β_i 's are the dual multipliers associated with the weight restrictions (6). We discuss their role further below. When all α_i 's and β_i 's are zero, (P2) is the envelopment formulation of an output-orientated DEA model in the specific case where there are no inputs: a common expansion factor φ is applied to all sub-indicators for the assessed observation, such that this expanded observation is feasible in the sense of being a linear combination of observed 'benchmark' observations.¹⁰ Clearly, such information – particularly regarding benchmark observations, i.e. those observations j for which $\lambda_j > 0$ – is valuable for benchmarking and will be used in the next section to identify cities' benchmarks and performance targets.

4. Results

4.1. Full-sample general findings

Table 5 presents the BoD scores and importance shares for the top 25 cities (out of 155), next to their standardised C3 Index scores and ranks. Five cities, from three population groups – Paris (XXL), Copenhagen (XL), Eindhoven (M), Linz (M) and Cork (M) – have the maximum BoD score of 1 (although in the latter two cases this is a rounding effect, with actual BoD scores of

variation. Clearly, this does not in itself mitigate the discrepancy problem as identified by Sarrico and Dyson (2004), as it could imply that the evaluated observation itself would violate the imposed bounds. On the other hand, and as pointed out by a referee, the approach of Zanella et al. (2015) eventually results in a single frontier, so avoiding the inefficient peer problem that we address in Section 4.

¹⁰ In fact, without the multipliers α_i and β_i , (P2) is one of the earliest 'outputs only' DEA models, going back to Lovell, Pastor and Turner (1995). For completeness, we point out that the convexity constraint $\sum \lambda_j = 1$ in (P2) is a weak inequality ($\sum \lambda_j \leq 1$) when taking the exact dual of (P1). However, as shown by Lovell & Pastor (1999), this will always be binding in an optimal solution to (P2), so explaining its *prima facie* appearance as a variable returns to scale output-oriented DEA model.

Table 5
BoD results for the top twenty-five cities in the CCCM.

City	BoD score	C3 Index score 2017	C3 Index rank	D1.1share	D1.2Share	D2.1share	D2.2share	D2.3share	D3.1share	D3.2share	D3.3share	D3.4 share
Paris	1.00	64.2	1	42.2%	35.1%	15.3%	0.7%	0.6%	0.4%	5.3%	0.4%	0.05%
Copenhagen	1.00	50.9	2	33.5%	35.2%	5.6%	0.7%	0.6%	0.4%	23.7%	0.4%	0.05%
Eindhoven	1.00	50.5	3	42.2%	24.2%	3.2%	18.1%	0.6%	0.4%	11.0%	0.4%	0.05%
Linz	1.00	45.3	5	35.6%	7.2%	33.1%	0.7%	0.6%	0.4%	22.1%	0.4%	0.05%
Cork	1.00	41.3	12	42.2%	17.8%	3.2%	0.7%	0.6%	7.1%	23.7%	0.4%	4.41%
Dublin	0.97	43.1	8	42.2%	7.2%	5.8%	0.7%	2.4%	6.1%	23.7%	7.5%	4.41%
's-Hertog.	0.97	43.0	9	38.7%	35.2%	3.3%	0.7%	0.6%	0.4%	9.3%	7.5%	4.41%
Galway	0.96	37.4	18	42.2%	7.2%	3.2%	0.7%	0.6%	17.8%	23.7%	0.4%	4.41%
Stockholm	0.94	43.1	7	6.5%	21.6%	14.9%	20.4%	0.6%	0.4%	23.7%	7.5%	4.41%
Stuttgart	0.89	42.2	11	6.5%	7.2%	29.3%	20.4%	0.6%	0.4%	23.7%	7.5%	4.41%
Florence	0.87	36.1	27	42.2%	35.2%	3.1%	0.7%	0.6%	0.4%	11.9%	5.9%	0.05%
Norwich	0.86	35.5	16	42.2%	7.2%	3.2%	0.7%	0.6%	10.6%	23.7%	7.5%	4.41%
Lund	0.86	39.2	32	6.5%	29.1%	7.4%	20.4%	0.6%	0.4%	23.7%	7.5%	4.41%
Munich	0.85	42.7	4	6.5%	7.2%	29.3%	20.4%	0.6%	0.4%	23.7%	7.5%	4.41%
Amsterdam	0.85	46.5	10	42.2%	7.2%	3.2%	0.7%	12.5%	6.8%	15.6%	7.5%	4.41%
Lisbon	0.83	43.6	6	42.2%	7.2%	3.2%	0.7%	25.8%	0.4%	20.2%	0.4%	0.05%
Limerick	0.83	32.3	45	42.2%	7.2%	3.1%	0.7%	0.6%	17.7%	23.7%	0.4%	4.41%
Heidelberg	0.81	39.4	15	6.5%	35.1%	3.1%	8.3%	0.6%	10.6%	23.7%	7.5%	4.41%
Helsinki	0.81	35.6	31	6.5%	7.2%	29.3%	20.4%	0.6%	0.4%	23.7%	7.5%	4.41%
Edinburgh	0.80	37.4	20	33.0%	7.2%	3.1%	0.7%	0.6%	21.3%	23.7%	6.0%	4.41%
Weimar	0.80	35.4	24	38.6%	35.1%	3.2%	1.1%	0.6%	0.4%	9.1%	7.5%	4.41%
Brussels	0.80	37.0	34	7.2%	7.2%	3.1%	0.7%	45.8%	0.4%	23.7%	7.5%	4.41%
Prague	0.79	39.4	13	42.2%	7.2%	3.1%	0.7%	27.1%	0.4%	18.9%	0.4%	0.05%
Frankfurt	0.78	37.8	17	6.5%	23.1%	33.1%	0.7%	0.6%	0.4%	23.7%	7.5%	4.41%
Utrecht	0.78	37.1	22	6.5%	7.2%	28.1%	0.7%	0.6%	21.3%	23.7%	7.5%	4.41%

Note: To avoid zero scores in the dimensions that would bring the shares to zero we have added 1 point to the 'original scores' of the dimensions. Hence, all C3 Index scores are 1 point above the 'original' index scores.

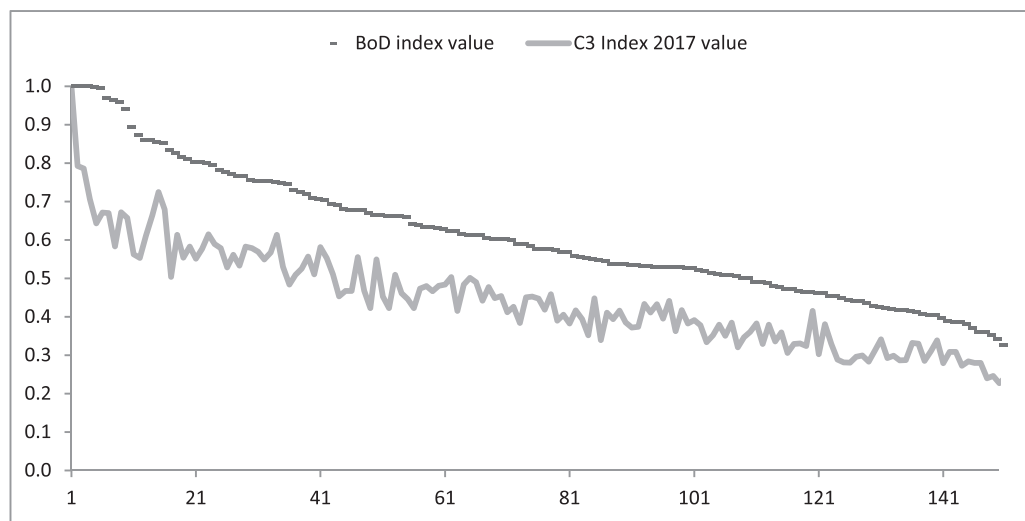


Fig. 1. C3 Index 2017 scores and DEA 'distance to the efficient frontier' scores for 155 cities.

Note: The original C3 Index scores have been rescaled (i.e. ratio of the C3 Index score over the maximum score in the sample) in order for the two metrics to be put on a comparable scale.

0.997 and 0.996 respectively). In the three 'true' cases with maximal scores, there are in fact multiple optimal solutions to (P1), and the resulting dimension importance shares of Paris, Copenhagen and Eindhoven as reported in Table 5 are, therefore, non-unique. For all other cities, there is no such concern. As expected, all importance shares are in accordance with the starting point of granting leeway to each city when assigning shares, while not violating the upper and lower bounds (6). While in each case at least one of the bounds is shown to be binding, the calculated importance shares are quite diverse, reflecting the differences in local relative strengths and weaknesses.

With the exception of Paris – which obtains the maximum score of 1 when rescaling the C3 Index scores on a 0–1 range – all the other cities in the sample improve their performance when

BoD is applied, albeit to a different extent (Fig. 1). The average improvement amounts to almost 16 percentage points, with the highest improvements amounting to more than 30 percentage points (e.g. Galway, Cork, Limerick, Florence, Norwich, 's-Hertogenbosch and Dublin). Cities that are modestly affected by introducing this flexibility in the weights are located in Central and Eastern Europe (e.g. Sofia, Sibiu, Plovdiv and Katowice). Improvement in these cases still amounts to 4–8 percentage points.

Fig. 1 further reveals that the results obtained using the BoD model (P1) are, in general, highly consistent with the C3 Index 2017 scores. This consistency is further underscored by the very high degree of correlation between the C3 Index scores and the BoD scores (i.e. the Pearson score correlation coefficient is 0.949; the Spearman rank correlation is 0.964), which is somewhat to be

Table 6
Paris as an 'influential' observation.

'Top' City (maximum BoD score)	Super-efficiency	Average ratio of BoD score in full model and BoD score in model without 'Top' city	# acting as peer for other obs.	Proportion of cases where observation accounts for $\geq 90\%$ (100%) of $\Sigma \lambda$ for other observations
Paris	1.479	0.900	148	49.4% (41.6%)
Copenhagen	1.053	0.996	68	3.2% (1.3%)
Eindhoven	1.060	0.999	9	0% (0%)

expected considering that, while allowing for some variations, we have respected essential parts of the C3 Index's weight structure as proposed by the experts.¹¹

4.2. Identifying benchmark cities and targets based on city clusters

We turn next to the benchmarking information provided by the BoD scores. The identification of benchmarks should be done with caution, especially when treating such a diversified sample of EU cities. In this respect, it should particularly be recalled that the original C3 Index distinguished cities based on different socioeconomic and demographic characteristics, with the objective of avoiding unrealistic comparisons.

The general results in Table 5 and Fig. 1 do not account for this desirable feature. In this respect, a first thing to note is that only 3 out of 155 cities get the maximum BoD score of 1, raising the concern that Paris, Copenhagen and Eindhoven might be outliers and are, in that sense, not the most appropriate (credible) benchmarks for many of the other cities in the sample. In fact, there is quite some evidence that Paris does have a considerable influence on the BoD scores of the other cities. Table 6 below reports some statistics that have been suggested in the literature (e.g. Pastor, Ruiz & Sirvent, 1999; Tran, Shively & Preckel, 2010) to detect outlier/influential behaviour. In the table we report the three cities' superefficiency score (Andersen & Petersen, 1993), next to the average ratio of each city's BoD score in a model with and without the potential influential observation. As far as the latter is concerned, we note additionally that, for 115 cities, the BoD score increases by at least 5% as soon as Paris is removed from the sample. For a smaller sample of 82 cities, the BoD score would even increase by at least 10%, once Paris is removed.

The last two columns of Table 6 relate to information on the three cities' capacity as benchmarks, and are retrieved from the dual ('envelopment') formulation (P2).

A first statistic counts in how many of the 154 cases (i.e. 155 minus the scenario in which the city itself is assessed) Paris, Copenhagen or Eindhoven obtain a strictly positive λ -value in (P2), indicating how often they are identified as a benchmark for other cities. Paris is again identified as highly influential ($\lambda_{\text{Paris}} > 0$ for more than 95% of all cities). The last column refers to the magnitude of the λ -values, indicating how much a city contributes to

Table 7

Global Frontier Difference Indices and cluster/pooled frontier BoD ratios for 155 cities.

GFI (cluster B, cluster A)	Average ratio of cluster-specific and 'pooled' BoD score		
(Cluster 2, Cluster 1)	1.19	Cluster 1	0.99
(Cluster 3, Cluster 1)	1.83	Cluster 2	0.87
(Cluster 3, Cluster 2)	1.54	Cluster 3	0.60

identification of the dimension values that serve as a benchmark for other cities. Again, Paris has an important impact as global benchmark for many cities: in about half of the cases we observe $\lambda_{\text{Paris}} > 0.9$ (where, to recall from (P2), $\Sigma \lambda = 1$), and in more than 4 out of 10 cases in which Paris is identified as a benchmark, the French capital is the only city that is endogenously selected as another city's benchmark.

There is also a significant performance gap between Paris and all other cities in terms of the original C3 Index scores (see Table 1 where the C3 Index score for Paris is over 25% higher than that of the second-ranked city, Copenhagen). This is explained by the fact that Paris is a Top 5 or even best performer for several dimensions. All these observations suggest that Paris is not a suitable benchmark for the vast majority of the cities analysed.

Therefore, building further on the idea of comparing peer cities as is done in the CCCM, we test whether notable differences can be identified between different typologies of cities. To this end, we use the demographic and socioeconomic characteristics reported in Table 2, although we do not consider them separately. Instead, we identify groups of cities that have similar population sizes, employment rates and income levels, using cluster analysis. Ward's method allows us to identify three distinct clusters of similar size (i.e. 47, 63 and 45 cities in cluster 1, 2 and 3 respectively, corresponding to 30%, 41% and 29% of the 155 cities in the sample).¹²

In itself, this *ex ante* clustering of the 155 cities into three groups does not provide any evidence of possible systematic differences between attainable best practices. We therefore compare such differences by calculating both 'Global Frontier Difference Indices' (Asmild & Tam, 2007; Van Puyenbroeck & Rogge, 2018) and 'between cluster' aggregate scores (Karagiannis & Karagiannis, 2018).

The left part of Table 7 (GFI (cluster B, cluster A)) reports results for the former, where we use the formula

$$GFI(cl_B, cl_A) = \left[\prod_{j=1}^{155} (CI_j^{cl_B} / CI_j^{cl_A}) \right]^{1/155}$$

i.e. we calculate the BoD scores $CI_j^{cl_B}$ for all $j = 1, \dots, 155$ cities relative to a best practice frontier only constituted by cities belonging to cluster 'cl_B' and divide these by their corresponding index scores $CI_j^{cl_A}$ in which only cities belonging to cluster 'cl_A' are eligible to serve as benchmarks. A GFI value higher than 1 implies that, on average, $CI_j^{cl_A} < CI_j^{cl_B}$, and hence that the best-practice frontier constituted by cluster *a*-cities is more demanding — and in that sense

¹¹ Similar findings are obtained if we narrow down the range of possible importance shares in (6). In an alternative model, we allowed these to vary between the 10th and 90th percentile value of the implied importance share data, rather than between the minimum and maximum importance shares as shown in Table 4. This entails, for example, that D1.1's relative contribution to a city's total score is at least 10.2% and at most 30.1%, that of D1.2 is at least 15.2% and at most 28.9%, etc. The Pearson correlation with the C3 Index score for this alternative model is 0.949, and the rank correlation equals 0.942. When comparing this model with the BoD model as used in the main text, these figures are 0.916 and 0.924 respectively. An obvious downside of this additional narrowing is that for some cities (namely 37) the narrower bounds come into conflict with these cities' own implicit importance shares (5). Conversely, if we allow for a fully unrestricted BoD model, the correlations of the scores with either the C3 Index (0.769) or the BoD scores for our restricted model in the main text (0.790) are still high, but lower. The results of such an unrestricted BoD model also show, however, the prevalence of the zero-weight problem discussed in the main text: on average across the cities, 5.7 out of 9 dimensions get a zero weight.

¹² The full list of cities assigned to the three clusters is available upon request.

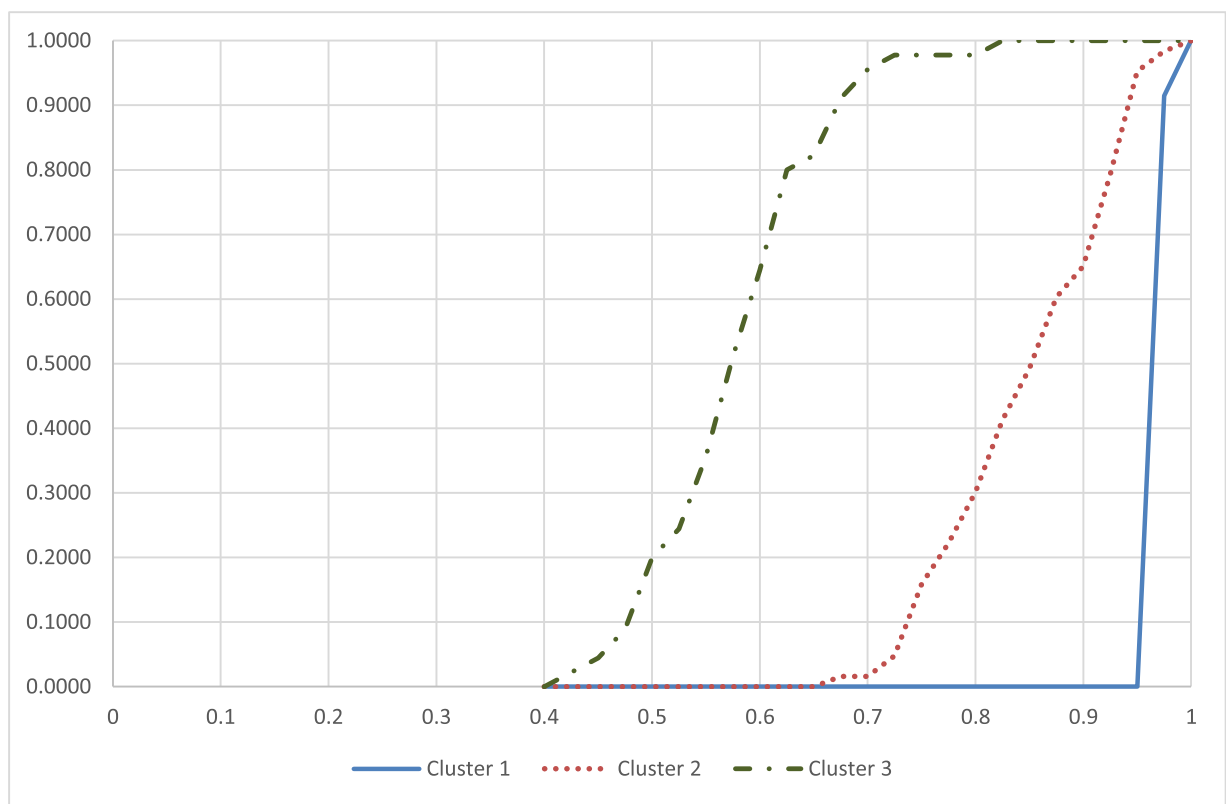


Fig. 2. Cumulative distribution of metafrontier ratios for three clusters of 155 cities.

outperforming — cluster *b*'s best practices. Cluster 1 clearly outperforms Cluster 2 and Cluster 3, and Cluster 2 outperforms Cluster 3. The right part of Table 7 (Average ratio of cluster-specific and 'pooled' BoD score) provides similar information. However, here we compare BoD index scores, as obtained when benchmarking only within the cluster to which a specific city belongs, with scores resulting from a comparison with the full 'metafrontier'. The closer this (average) ratio is to 1, the more a cluster frontier coincides with the pooled frontier (or 'metafrontier'). This is clearly the case for Cluster 1, whereas the Cluster 3 cities have a frontier rather far removed from the metafrontier.

The clear dominance of Cluster 1, which amounts to a first-order dominance of Cluster 1 over Cluster 3, is further confirmed by the cumulative distribution of these metafrontier ratios, as shown in Fig. 2.

4.3. Importance-consistent benchmarks

We illustrate how fit-for-purpose policy insights can be drawn through the identification of suitable benchmarks and targets for three cities selected from Cluster 2: Bilbao (Spain), Umeå (Sweden) and Kraków (Poland). We single out these cities as they have all invested considerably in positioning themselves as cultural and creative cities over the past few decades, albeit pursuing different policy objectives. Bilbao has mostly invested in culture as part of an ambitious economic and urban regeneration plan aimed at overcoming the industrial crisis of the 1970s (Plaza, Tironi & Haarich, 2009). Umeå has instead focused on 'the importance of cultural life for people's health, creativity and personal development' (City of Umeå, 2010), first as part of its European Capital of Culture (ECoC) project in 2014, and then with policy actions aimed at further fostering the culture–wellbeing link. Finally, Kraków has invested in the revitalisation of local cultural highlights and attractions, with a view to fostering tourism and invigorating the creative sector. It is

Table 8
'Possible' benchmarks for Bilbao, Kraków and Umeå (before consistency check) within Cluster 2 (63 cities).

City	BoD score (within cluster)	Identified benchmarks before consistency check ('λ' –value indicates% contribution of peer city to benchmark)	
Bilbao	0.577	Cork (λ=23.7%) Linz (λ=64.3%)	Eindhoven (λ=12.0%)
Kraków	0.734	Eindhoven (λ=20.2%)	Lisbon (λ=79.8%)
Umeå	0.913	Cork (λ=18.8%) Linz (λ=16.3%)	Eindhoven (λ=15.8%) Glasgow (λ=49.1%)

therefore relevant to see where these cities stand and where they can improve their performance by directing future policy action. At the same time, their achievements and identified targets could inspire other cities in Europe sharing similar priorities and socio-economic characteristics. Indeed, Cluster 2 groups the highest number of cities in the sample compared to the other two clusters (63 in Cluster 2 vs 47 in Cluster 1 and 44 in Cluster 3), gathering cities mostly of medium-size (250,000–500,000 inhabitants).

For a global analysis of Cluster 2 cities, the weights are allowed to vary within the lower and upper bounds identified in Table 8. Their BoD score (within cluster) and possible benchmarking cities are listed in the table below.

Note that we label these benchmarks as 'possible'. Here we encounter the problem, already signalled earlier, that by imposing the weight restrictions on each evaluated city only, other cities (including the identified peers) will not necessarily behave in a way that is consistent with these weight restrictions. Take, for instance, the case of Kraków. Table 9 reports the optimal weights for Kraków (first line) as resulting from applying (P1) within Cluster 2, and then applies these weights to the dimension scores for the two peers identified, Eindhoven and Lisbon. This generates the importance shares $(w_{i,Kraków}y_{i,Eindhoven})/\sum w_{i,Kraków}y_{i,Eindhoven}$ resp.

Table 9

Kraków's optimal weights and importance shares for Eindhoven and Lisbon, along the nine dimensions of the Cultural and Creative Cities Monitor framework.

	D1.1 Cultural Venues & Facilities	D1.2 Cultural Participation & Attractiveness	D2.1 Creative & Knowledge-based Jobs	D2.2 Intellectual Property & Innovation	D2.3 New Jobs in Creative Sectors	D3.1 Human Capital & Education	D3.2 Openness, Tolerance & Trust	D3.3 Local & International Connections	D3.4 Quality of Governance
Kraków optimal weights	0.00205	0.00641	0.00156	0.00028	0.00392	0.00508	0.00144	0.00051	0.00061
Eindhoven importance shares	11.14%	<u>35.25%</u>	7.15%	2.87%	11.43%	18.46%	4.15%	4.11%	<u>5.43%</u>
Lisbon importance shares	13.15%	29.83%	4.35%	<u>0.14%</u>	24.77%	19.00%	3.66%	1.48%	3.61%

$(w_{i,Kraków}y_{i,Lisbon}) / \sum w_{i,Kraków}y_{i,Lisbon}$. In both cases, there are instances in which importance bounds are violated if the only additional weight restrictions are of the type in Eq. (6), i.e. imposed on the evaluated city only and not on others (including any peers). These are the figures underlined in Table 9.

This discrepancy problem is also manifested in an arguably more worrying variant. In Table 8, cities such as Cork, Linz and even Lisbon are, next to Eindhoven, indicated as potential benchmarks, which fits with their own maximum BoD score of 1 when the benchmarking exercise is limited to the subsample of Cluster 2 cities rather than the full sample (where these three cities had lower scores – see Table 5 earlier). Glasgow, on the other hand, does not obtain the maximum score when the weight restrictions apply to Glasgow itself, not even within Cluster 2 (see Table A1 in Appendix 1), but this changes when these restrictions are imposed on another city, as demonstrated by the case of Umeå in Table 8. In other words, using Umeå's most favourable weights actually yields a higher score for Glasgow than when this exercise is tailored to Glasgow itself, thus fully contradicting the core element of the underlying 'benefit-of-the-doubt' philosophy. Cases such as this clearly underscore the perils of virtual output restrictions when applied to the evaluated observation only.

As stated above, while this risk was already identified at the theoretical level by Sarrico and Dyson (2004), this issue has been largely sidestepped in BoD applications. These authors also discussed the seemingly compelling solution of appending a programme such as (P1) with additional virtual output restrictions applying to all observations in the sample, but noted its propensity to lead to unfeasible solutions.¹³ Here, we suggest dealing with these inconsistencies using an intermediate way, by ensuring that the restrictions (6) are imposed not only on the evaluated city itself, but also on the benchmarking cities as identified in a first round.

The latter can be done in several different ways. A first option is to impose the weight constraints (6) both to the evaluated city itself and to each city that is identified as a possible benchmark for that evaluated city. However, as Table 8 shows, proceeding in a similar way for all cities would require changing (extra) weight constraints depending on the specific city that is evaluated. Furthermore, it must be noted that this procedure may, on some occasions, lead to an iterative process in which several rounds are needed before one is left only with truly weight-consistent targets.

We have therefore considered a stricter application of the consistency idea, by appending the additional consistency restrictions to all cities that have been identified as potential benchmarks in the first analysis (i.e. cities in Table A1 in the online Appendix I with a BoD score of 1 or 0.99). In other words: regardless of the specific city being evaluated, we ensure 'importance consistency' for Cork, Eindhoven, Galway, Glasgow, Heidelberg, Leuven, Linz, Lisbon, Manchester, 's-Hertogenbosch and Umeå. For the three cities under consideration, the results are summarised in Table 10.

Table 10

Identified benchmarks for Bilbao, Kraków and Umeå (after consistency check).

City	BoD score (within cluster)	Identified benchmarks (after consistency check) (λ -value indicates % contribution to benchmark)
Bilbao	0.520	Eindhoven ($\lambda=34.7\%$) Linz ($\lambda=65.3\%$)
Kraków	0.637	Eindhoven ($\lambda=100\%$)
Umeå	0.808	Eindhoven ($\lambda=100\%$)

As this method implies a considerable addition of weight restrictions (i.e. 11 cities $\times$ 2 inequalities $\times$ 9 indicators in our Cluster 2 sample), one should expect significant drops in the eventual BoD scores. A full list of scores generated in this fashion is found in Table A1 in Appendix I. On average, scores drop by 0.067 (or 9.5%) on adding the 198 additional weight restrictions. The maximal drop is observed for York, with a '100%' BoD score going down to 82.17% after ensuring full consistency with the 11 possible benchmarks.¹⁴ For 7 out of the 63 cities, the index score drops by less than 5%; 27 others have a downward effect on their score between 5 and 10%; and for 24 others the score drop ranges between 10 and 15%.¹⁵

4.4. Targets and relative strengths

We briefly illustrate how the identification of (importance-consistent) benchmarks further results in identifying targets for a city. Table 11 shows the actual dimension data for Bilbao.

Eindhoven and Linz are identified as Bilbao's peers by the BoD programme, meaning that they are the two cities that achieve a maximum score of 1 with the optimum weight structure identified for Bilbao. These benchmarks are therefore relevant for Bilbao for learning purposes from a policy development perspective.

The targets that Bilbao should aim to achieve can be obtained by a weighted combination of the values observed in Eindhoven and Linz. However, recalling (P2), it should be pointed out that Bilbao's target values as reported in Table 11 do not coincide with 'proportional expansion' targets. Such proportional expansion, captured by the value of φ in the objective function of (P2), would be the result generated by a BoD programme without additional weight restrictions. But, as is apparent in (P2), appending further weight restrictions in fact implies introducing additional, dimension-specific, corrections through the dual multipliers (the α_i 's and β_i 's). To provide a specific example, Bilbao is relatively good on Quality of Governance (D3.4), to the extent that proportionally inflating its actual value would imply that the (radially, φ -expanded) value so obtained ($70.8/0.520 = 136$) violates the upper importance bound for this dimension. The binding upper bound entails a positive (β_i) value for this dimension, which in turn en-

¹³ Indeed, even when confined to our subsample of 63 Cluster 2 cities, this would mean that 1116 additional weight restrictions (capturing a lower and an upper bound restriction for the nine dimensions for each of the 62 other cities) should be added to the BoD linear programme.

¹⁴ But note that York is a peculiar case in the sense that it is efficient 'by default' in the base case: it never appears as a possible target for any other city in Cluster 2, thus indicating it has a dimension indicator composition that is quite distinct from that of the other cities. Intuitively, by appending the consistency restrictions with those of 11 other cities, York's atypical mix is then penalised, which shows up in the lower score.

¹⁵ A further analysis of what drives the score differences for different cities is provided in an online appendix (Appendix II).

Table 11
Scores and targets for Bilbao along the nine dimensions of the Cultural and Creative Cities Monitor framework.

	Domains	Actual Bilbao	Actual Eindhoven ($\lambda = 34.7\%$)	Actual Linz ($\lambda = 65.3\%$)	Targets for Bilbao
Cultural Vibrancy	D1.1 Cultural Venues & Facilities	17.64	54.35	55.28	54.95
	D1.2 Cultural Participation & Attractiveness	33.00	54.97	52.22	53.17
Creative Economy	D2.1 Creative & Knowledge-based Jobs	31.00	45.90	64.50	58.04
	D2.2 Intellectual Property & Innovation	8.70	101.00	40.00	61.16
	D2.3 New Jobs in Creative Sectors	11.80	29.13	17.30	21.40
Enabling Environment	D3.1 Human Capital & Education	15.03	36.37	18.02	24.38
	D3.2 Openness, Tolerance & Trust	30.02	28.78	50.33	42.85
	D3.3 Local & International Connections	17.77	80.37	41.07	54.70
	D3.4 Quality of Governance	70.80	89.00	83.30	85.28

Note: To avoid zero scores in the dimensions that would bring the shares to zero, in previous steps we have added 1 point to the ‘original scores’ of the dimensions. Hence, all C3 Index scores are 1 point above the ‘original’ index scores. This means that Eindhoven’s score on D2.2, for instance, should be intended as ‘100’ instead of ‘101’.

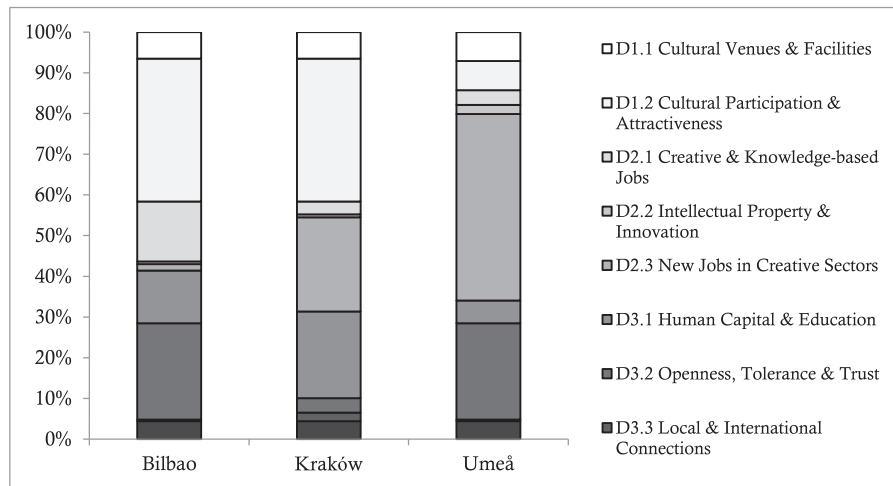


Fig. 3. BoD importance shares for the nine dimensions in the Cultural and Creative Cities Monitor framework.

sures that the target value as identified in (8) is adjusted downwards (to 85.28, as can be seen in the table).

Overall, Bilbao could potentially improve its performance across all the policy dimensions identified, taking inspiration from the identified peers. Both Eindhoven and Linz are two successful examples of industrial centres that have been able to re-shape their economies and image by investing in creativity, new technologies and R&D as well as by setting up a lively cultural offer. Bilbao is pursuing a very similar path: the city has committed itself to finding a new economic model for the future, building on its manufacturing tradition and at the same time creating better opportunities for residents and visitors in terms of leisure and culture (e.g. the Guggenheim Museum Bilbao has become a truly global icon). Its culture-led development strategy also aims at strengthening the economic dimension of the arts and culture with a view ‘to allow the city to be competitive in sectors with a greater future forecast’ (Otaolea, 2014). Coherent with this view, the Zorrotzaurre project – a major urban regeneration project currently being implemented– aims to turn a former industrial area, very close to the city centre, into a ‘creative island’ to support creative entrepreneurship.¹⁶ Successful practices from Eindhoven and Linz¹⁷

could help Bilbao to shape this new area and policy actions for performance improvement.

In addition to the identification of benchmarks and targets, another useful piece of information that can be retrieved from the BoD approach relates to the optimal weight structure itself and the associated dimension importance identified for each city. This information serves to identify relative strengths and areas to be improved for each city.

Fig. 3 presents this information for the three cities selected from Cluster 2. It can be seen that the optimal weight structure obtained under the BoD scenario represents a relatively balanced distribution of weights for the different dimensions, which is the consequence of the additional weight restrictions imposed. Overall, of course, we also observe some different weight structures, suggesting that the three cities show relative strengths in different areas.

The BoD results for Bilbao confirm the city’s efforts over the last two decades to invest in ‘Cultural Vibrancy’: the first two dimensions alone account for more than 40% of its total BoD score, of which slightly more than 35% is to be attributed to D1.2 Cultural Participation and Attractiveness, while ‘Creative Economy’ dimensions contribute around 17% of the total score. ‘Enabling Environment’ dimensions again contribute about 40% of Bilbao’s aggregate performance score, of which about 24% is represented by dimen-

¹⁶ <http://www.zorrotzaurre.com/en/>.

¹⁷ Eindhoven, for instance, has been able to develop a lively high-tech environment, is home to the Design Academy and hosts a vibrant start-up scene – a major working axis for Bilbao and the entire Basque Country region. In the past two decades, Linz too has considerably invested in the promotion and growth of cultural and creative sectors and professionals. In addition to having been European Capital of Culture in 2009, Linz has obtained the title of UNESCO City of Media Arts due to the city’s significant investment in media arts and digital competences, through initiatives such as the annual Ars Electronica Festival and Prix Ars Electronica as well as Ars Electronica Centre and Futurelab. Also, the former tobacco factory Tabakfab-

rik today plays an important role in the international positioning of Linz. Developed by the city in cooperation with Creative Region and Upper Austria’s creative industries development company, the Tabakfabrik Linz has established itself as a focal point for cultural and creative start-ups. Linz is in fact very strong on D2.1 Creative & Knowledge-based Jobs, although Eindhoven seems to foster better dynamics as regards innovation (D2.2) and job creation (D2.3).

sion D3.2 Openness, Tolerance & Trust. The weight structure thus corroborates the need to further pursue the city's investment plans aimed at creating a new specialisation in the field of creative entrepreneurship and innovation.

Kraków has instead managed to keep a more balanced weight structure, although the 'Cultural Vibrancy' dimensions remain prominent (around 40%) and very much in line with the city's efforts in revitalising the local cultural institutions since the ECoC project in 2000. The creation of the Kraków Festival Office on the occasion of Kraków 2000, for instance, is a very important legacy of the ECoC title as it continues to produce, coordinate and support Kraków's leading cultural institutions and events. The contribution of job creation in the cultural and creative sectors to the total C3 Index score is also noteworthy (accounting for around 23%). This job creation is likely influenced by the ability of the Eastern regions overall to catch up since the 1990s, due to their capacity to attract foreign direct investment but also to the presence of human capital (in Kraków accounting for a considerable 21% of the total score) (Tondl & Vuksic, 2003).

Umeå depicts a quite different situation, with nearly 46% of its total BoD score generated by dimension D2.3 New Jobs in Creative Sectors alone. Umeå has indeed experienced a rapid growth in the last decade, becoming one of the fastest growing cities in Europe (City of Umeå, 2014). The city is also relatively strong on dimension D3.2 Openness, Tolerance & Trust (around 24% of its C3 Index score), a feature that it is not uncommon in most Northern European cities. Umeå's challenge is to further promote the 'Cultural Vibrancy' dimension as a new specialisation area. In fact, this ambition is established in the city's cultural strategy, which aims both to revitalise its cultural activities and to strengthen cultural participation as a driver of personal wellbeing. Such evidence comes at a timely moment for Umeå, as it further supports the definition of the objectives and priorities of the current cultural strategy, which is currently under revision.

5. Conclusion

The recognition of diversity and the careful selection of relevant peers are two fundamental considerations in the European Commission's Cultural and Creative Cities Monitor. In this paper, we uphold the view that these considerations justify complementing the existing C3 Index with a Benefit-of-the-Doubt model to evaluate the aggregate performance of the cultural and creative economy in European cities. This model allows us to overcome the limitations of an index with fixed weights, and at the same time identifies the relative strengths and areas to be improved for each city. In particular, this approach allows for a 'fairer' comparison between cities by respecting their different local contexts and policy choices. This is consistent with a more flexible, inclusive and non-coercive approach to culture-led development, as defended in the relevant literature (e.g. Smith & Warfield, 2007).

At the same time, the flexibility we introduce in a city's weighting scheme is not unlimited, as we keep building on existing expert opinion to define restrictions on a city's range of allowable weights. A novel aspect of this paper revolves around the way this expert opinion is translated into weight restrictions in the BoD model. Both the composite indicator literature in general, as well as the DEA (BoD) literature in particular, make an explicit distinction between 'pure weights' in an index on the one hand and the 'importance' of a constituent sub-indicator for the total score on the other. While in empirical applications the theoretical distinction between these two is usually blurred, it is precisely by building on this distinction that we can join a set of expert weights that is common to all cities with city-specific restrictions on dimension importance shares in the final composite indicator.

In our application, we opted for considerable latitude in the ranges of allowable importance shares, by anchoring the bounds on the sample minima and maxima for the respective dimension scores. While directly led by our concern to both illustrate and respect the transition from stated weights to implied importance shares for all cities, it is conceivable that the resulting dimensions' shares may in some specific instances be conceived as being too far from what was initially suggested by experts. Different approaches, e.g. discarding the bottom and top 5% of the sample dimension data when defining the importance bounds, are certainly conceivable. While such considerations are eventually application-specific, future research could specifically address this concern.

In the original monitoring exercise, the crucial tenet of selecting appropriate benchmarks is mainly accomplished by the *a priori* definition of city types, depending on an exogenous characteristic. Neglecting this tenet in a BoD model seems ill-advised. The dominant performance of Paris is of particular concern as it would entail a dominance of this metropolis as 'benchmark' even for considerably smaller cities. We thus preserve the gist of the 'city type' idea in that we retain the same characteristics to distinguish between cities on an *ex ante* basis, as done in the Cultural and Creative Cities Monitor. In technical terms, the underlying idea is that differences in exogenous characteristics may instigate systematic differences in terms of the best practices that specific observations can attain. However, we depart from the original classification and look at population, GDP per capita and employment rates simultaneously, using a cluster analysis to endogenise the city grouping process. In other words, we complement the *ex ante* classification with an *ex post* check that confirms the presence of performance differences between cities. We note that other approaches could have been used in this respect, and indeed provide an obvious avenue for further research. In particular, while we have opted in this paper for a purely deterministic approach, Monte Carlo-based techniques (e.g. conditional order-m models, see e.g. Verscheide & Rogge, 2012) provide an alternative way to assess the systematic impact of exogenous variables on cities' performance.

We have also argued that selecting appropriate benchmarks also necessitates guaranteeing that a city's endogenously identified peers should adhere to the very same importance bounds as those applying to the evaluated city itself. Accordingly, we have proposed a way to deal with the potential ambiguity in this respect that was originally identified by Sarrico and Dyson (2004). We recall that the specific approach we adopted above – ensuring that virtual output restrictions hold for the assessed observations and for all observations that have been identified as potential benchmarks for the full sample prior to a further consistency check – is not the only possibility. One could even argue that this procedure is too strict, as it entails that in most cases weight restrictions are also imposed unnecessarily on cities that are not a possible benchmark for the evaluated city. The advantage of using it, however, is that it clearly avoids setting up *ad hoc* comparisons for each city separately. Both approaches are likely to generate somewhat different results, but it goes beyond the scope of the current paper to address this observation, and the relative merits of these (and possibly other) approaches, in a systematic manner.¹⁸

By using a flexible structure of weights for dimensions in an indicator framework, we show that it is possible to identify relevant benchmarks and targets to improve the current performance of the cities analysed. The three cities that we singled out for fur-

¹⁸ To give an idea about resulting differences: applying weight restrictions on city-specific 'potential benchmarks' would lead to a BoD-score for Bilbao of 0.571, for Kraków of 0.727, and for Umeå of 0.861. The fact that these numbers are higher than those appearing in Table 10 is of course a straightforward consequence of the lower number of extra weight restrictions when only city-specific rather than sample-wide potential benchmarks are considered.

ther analysis all have potential for improvement, but the appropriate pathways to their improvement are different (e.g. focus on cultural facilities and participation, or innovation and creative entrepreneurship). Furthermore, the results straightforwardly identify relative strengths and weaknesses of the cities, through investigation of the optimal weight structure for the nine dimensions in the indicator framework, which helps to identify (or support) new directions for future development, whether or not already identified by local development strategies. All in all, the proposed toolkit contributes to guiding EU cities towards formulation and monitoring of more effective policy measures to foster culture-led development.

Declaration of Competing Interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.ejor.2020.05.058](https://doi.org/10.1016/j.ejor.2020.05.058).

References

- Andersen, P., & Petersen, N. C. (1993). A procedure for ranking efficient units in data envelopment analysis. *Management Science*. <https://doi.org/10.1287/mnsc.39.10.1261>.
- Asmild, M., & Tam, F. (2007). Estimating global frontier shifts and global Malmquist indices. *Journal of Productivity Analysis*, 27(2), 137–148. <https://doi.org/10.1007/s11123-006-0028-0>.
- Bakhshi, H., MacVittie, E., & Simmie, J. (2008). Creating innovation. Do the creative industries support innovation in the wider economy. *NESTA*. <https://doi.org/10.1108/EUM000000000006035>.
- Becker, W., Saisana, M., Paruolo, P., & Vandecasteele, I. (2017). Weights and importance in composite indicators: Closing the gap. *Ecological Indicators*, 80, 12–22. <https://doi.org/10.1016/j.ecolind.2017.03.056>.
- Blessi, G. T., Grossi, E., Sacco, P. L., Pieretti, G., & Ferilli, G. (2016). The contribution of cultural participation to urban well-being. A comparative study in Bolzano/Bozen and Siracusa, Italy. *Cities (London, England)*, 50, 216–226. <https://doi.org/10.1016/j.cities.2015.10.009>.
- Brousseau, D., Occelli, F., Genin, M., Cuny, D., Deram, A., & Lanier, C. (2020). Spatialized composite indices to evaluate environmental health inequalities: Meeting the challenge of selecting relevant variables. *Ecological Indicators*, 111, Article 106023. <https://doi.org/10.1016/j.ecolind.2019.106023>.
- Cherchye, L., Moesen, W., Rogge, N., Van Puyenbroeck, T., Saisana, M., Saltelli, A., et al. (2008). Creating composite indicators with DEA and robustness analysis: The case of the Technology Achievement Index. *Journal of the Operational Research Society*, 59(2), 239–251. <https://doi.org/10.1016/j.jors.2006.04.045>.
- Cherchye, Laurens, Moesen, W., Rogge, N., & Puyenbroeck, T. Van (2007). An Introduction to 'Benefit of the Doubt' Composite Indicators. *Social Indicators Research*, 82(1), 111–145. <https://doi.org/10.1007/s11205-006-9029-7>.
- City of Umeå. (2010). Cultural Policy Programme for Umeå Municipality.
- City of Umeå. (2014). Report from one of Sweden's fastest growing cities. Retrieved from https://www.umea.se/download/18.52289d7013a4b49e9872afb/1361886527341/Basbroschyr_2011_ENG.pdf
- Cooper, W. W., Ruiz, J. L., & Sirvent, I. (2011). Choices and uses of DEA weights. *Handbook on Data Envelopment Analysis* (pp. 93–126). Springer. Retrieved from <http://cio.edu.umh.es/2017/07/10/cooper-w-w-ruiz-j-l-sirvent-i-2011-choices-and-uses-of-dea-weights-in-william-w-cooper-lawrence-w-seiford-and-joe-zhueds-handbook-on-data-envelopment-analysis-nueva-york-spril/>.
- El Gibari, S., Gómez, T., & Ruiz, F. (2019). Building composite indicators using multicriteria methods: A review. *Journal of Business Economics*, 89(1), 1–24. <https://doi.org/10.1007/s11573-018-0902-z>.
- European Commission. (2010a). Europe, the world's No 1 tourist destination – a new political framework for tourism in Europe COM(2010) 352 final.
- European Commission. (2010b). Green Paper Unlocking the potential of cultural and creative industries COM(2010) 183. Retrieved from http://ec.europa.eu/commission_barroso/president/pdf/press_20090903_EN.pdf
- European Commission. (2010c). Regional Policy contributing to smart growth in Europe 2020 COM(2010) 553 final.
- European Commission. (2012). Promoting cultural and creative sectors for growth and jobs in the EU COM(2012) 537 final. Retrieved from <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52012DC0537&from=EN>
- European Commission. (2014a). For a European Industrial Renaissance COM(2014) 14 final.
- European Commission. (2014b). Towards an integrated approach to cultural heritage for Europe COM(2014) 477 final. Retrieved from www.culture-dev.eu/pdf/DeclarationofBrugesEN.pdf
- European Commission. (2016). Joint Communication to the European Parliament and the Council towards an EU strategy for international cultural relations JOIN(2016) 29 final.
- Evans, G., & Shaw, P. (2004). The contribution of culture to regeneration in the UK: A review of evidence – A report to the Department for Culture Media and Sport. Retrieved from <http://www.scholars-on-bilbao.info/fichas/EvansShaw2004.pdf>
- Ferilli, G., Sacco, P. L., Tavano Blessi, G., & Forbici, S. (2017). Power to the people: When culture works as a social catalyst in urban regeneration processes (and when it does not). *European Planning Studies*, 25(2), 241–258. <https://doi.org/10.1080/09654313.2016.1259397>.
- Gasser, P. (2020). A review on energy security indices to compare country performances. *Energy Policy*, 139, Article 111339. <https://doi.org/10.1016/j.enpol.2020.111339>.
- JRC-OECD (2008). Handbook on constructing composite indicators. In M. Nardo, M. Saisana, A. Saltelli, S. Tarantola, A. Hoffman, & E. Giovannini (Eds.), *Methodology and User Guide*. OECD Publishing ISBN 978-92-64-04345-9.
- Karagiannis, R., & Karagiannis, G. (2018). Intra- and inter-group composite indicators using the BoD model. *Socio-Economic Planning Sciences*. <https://doi.org/10.1016/j.seps.2017.01.002>.
- KEA. (2006). The Economy of Culture in Europe. *European commission*. <https://doi.org/10.1017/S0020859099000437>.
- Kwatra, S., Kumar, A., & Sharma, P. (2020). A critical review of studies related to construction and computation of Sustainable Development Indices. *Ecological Indicators*, 112, Article 106061. <https://doi.org/10.1016/j.ecolind.2019.106061>.
- Lovell, C. A. K., & Pastor, J. T. (1999). Radial DEA models without inputs or without outputs. *European Journal of Operational Research*, 118(1), 46–51. Retrieved from <http://cio.edu.umh.es/2018/01/16/lovell-c-a-k-pastor-j-t-1999-radial-dea-models-without-inputs-or-without-outputs-european-journal-of-operational-research/>.
- Miles, S., & Paddison, R. (2005). Introduction: The rise and rise of culture-led urban regeneration. *Urban Studies*, 42(5–6), 833–839. <https://doi.org/10.1080/00420980500107508>.
- Montalto, V., Tacao Moura, C.J., Langedijk, S., & Saisana, M. (2017). The Cultural and Creative Cities Monitor: 2017 edition. Retrieved from <http://publications.jrc.ec.europa.eu/repository/bitstream/JRC107331/kj0217787enn.pdf>
- Montalto, V., Tacao Moura, C. J., Langedijk, S., & Saisana, M. (2019). Culture counts: An empirical approach to measure the cultural and creative vitality of European cities. *Cities (London, England)*, 89, 167–185. <https://doi.org/10.1016/j.cities.2019.01.014>.
- OECD. (2009). The impact of culture on tourism. In. *OECD*. <https://doi.org/10.1787/9789264040731-en>.
- Otaola, I.B. (2014). Culture as the Engine of Bilbao's Economic and Social Transformation.
- Paruolo, P., Saisana, M., & Saltelli, A. (2013). Ratings and rankings: Voodoo or science. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 176(3), 609–634. <https://doi.org/10.1111/j.1467-985X.2012.01059.x>.
- Pastor, J. T., Ruiz, J. L., & Sirvent, I. (1999). Enhanced DEA Russell graph efficiency measure. *European Journal of Operational Research*. [https://doi.org/10.1016/S0377-2217\(98\)00098-8](https://doi.org/10.1016/S0377-2217(98)00098-8).
- Plaza, B., Tironi, M., & Haarich, S. N. (2009). Bilbao's Art Scene and the "Guggenheim effect" Revisited. *European Planning Studies*, 17(11), 1711–1729. <https://doi.org/10.1080/09654310903230806>.
- Potts, J. (2009). Why creative industries matter to economic evolution. *Economics of Innovation and New Technology*, 18(7), 663–673. <https://doi.org/10.1080/10438590802564592>.
- Pratt, A. C., & Jeffcutt, P. (2009a). Creativity, innovation and the cultural economy: Snake oil for the twenty-first century. *Creativity, innovation and the cultural economy* (pp. 3–19). <https://doi.org/10.4324/9780203880012>.
- Pratt, A. C., & Jeffcutt, P. (2009b). *Creativity, innovation and the cultural economy*. Routledge.
- Prior, J. H., & Blessi, G. T. (2012). Social Capital, Local Communities and Culture-led Urban Regeneration Processes: The Sydney Olympic Park Experience. *Cosmopolitan Civil Societies: An Interdisciplinary Journal*, 4(3), 78–96. <https://doi.org/10.5130/ccs.v4i3.2684>.
- Saisana, M., Saltelli, A., & Tarantola, S. (2005). Uncertainty and sensitivity analysis techniques as tools for the quality assessment of composite indicators. *Journal of the Royal Statistical Society. Series A: Statistics in Society*, 168(2), 307–323. <https://doi.org/10.1111/j.1467-985X.2005.00350.x>.
- Saltelli, A. (2007). Composite indicators between analysis and advocacy. *Social Indicators Research*, 81(1), 65–77. <https://doi.org/10.1007/s11205-006-0024-9>.
- Santagata, W. (2002). Cultural Districts, Property Rights and Sustainable Economic Growth. *International Journal of Urban and Regional Research*, 26(1), 9–23. <https://doi.org/10.1111/1468-2427.00360>.
- Sarrico, C. S., & Dyson, R. G. (2004). Restricting virtual weights in data envelopment analysis. *European Journal of Operational Research*, 159(1), 17–34. [https://doi.org/10.1016/S0377-2217\(03\)00402-8](https://doi.org/10.1016/S0377-2217(03)00402-8).
- Scott, A. J. (2001). Capitalism, cities, and the production of symbolic forms. *Transactions of the Institute of British Geographers*, 26(1), 11–23. <https://doi.org/10.1111/1475-5661.00003>.
- Smith, R., & Warfield, K. (2007). The Creative City: A Matter of Values. Retrieved from https://econpapers.repec.org/bookchap/elgeechap/12738_5f12.htm
- Throsby, D. (2001). *Economics and Culture*. Cambridge University Press, 194. <https://doi.org/10.1080/13547860306286>.
- Tondl, G., & Vuksic, G. (2003). What makes regions in Eastern Europe catching up?

- The role of foreign investment, human resources and geography. *ZEI Working Papers* Retrieved from <https://ideas.repec.org/p/zbw/zeiwps/b122003.html>.
- Tran, N. A., Shively, G., & Preckel, P. (2010). A new method for detecting outliers in Data Envelopment Analysis. *Applied Economics Letters*. <https://doi.org/10.1080/13504850701765119>.
- UNCTAD. (2010). Creative Economy Report 2010. In *English*. <https://doi.org/10.1007/s13398-014-0173-7.2>.
- UNCTAD. (2013). Creative Economy Report 2013 Special Edition. In *One United Nations Plaza*.
- van Oort, F. G. (2008). Philip Cooke and Luciana Lazzeretti (eds): Creative cities, cultural clusters and local economic development. *Journal of Cultural Economics*, 32(3), 237–240. <https://doi.org/10.1007/s10824-008-9070-2>.
- Van Puyenbroeck, T., & Rogge, N. (2018). Comparing regional human development using global frontier difference indices. *Socio-Economic Planning Sciences*. <https://doi.org/10.1016/j.seps.2018.10.014>.
- Verschelde, M., & Rogge, N. (2012). An environment-adjusted evaluation of citizen satisfaction with local police effectiveness: Evidence from a conditional data envelopment analysis approach. *European Journal of Operational Research*, 223(1), 214–225 Retrieved from <https://biblio.ugent.be/publication/2998026>.
- Wong, Y.-H. B., & Beasley, J. E. (1990). Restricting weight flexibility in data envelopment analysis. *The Journal of the Operational Research Society*, 41(9), 829. <https://doi.org/10.2307/2583498>.
- World Economic Forum. (2016). Factors for Enabling the Creative Economy. Retrieved from www.weforum.org
- Zanella, A., Camanho, A. S., & Dias, T. G. (2015). Undesirable outputs and weighting schemes in composite indicators based on data envelopment analysis. *European Journal of Operational Research*, 245, 517–530. <https://doi.org/10.1016/j.ejor.2015.03.036>.